

ST BENEDICT'S COLLEGE



SUBJECT
GRADE
EXAMINER
NAME
TEACHER

Mathematics Paper 1

11

Mr Benecke

DATE

MARKS

MODERATORS

DURATION

14+17 February 2017

70

Mrs Povall + Grade 11 teachers

80min

QUESTION NO	DESCRIPTION	MAXIMUM MARK	ACTUAL MARK
1	Equations and inequalities	36	
2	Simultaneous	9	
3	Nature of Roots	6	
4	Completing the Square	11	
5	Word Problem	8	
TOTAL		70	

QUESTION 1

36 MARKS

a) Solve for x :

i) $x^2 - 11x - 12 = 0$ (4)

$$(x + 1)(x - 12) = 0 \quad \checkmark \checkmark$$

$$x = -1 \quad \checkmark \text{ or } x = 12 \quad \checkmark$$

ii) $(x + 1)(x - 3) = 8$ (6)

$$x^2 - 2x - 3 = 8 \quad \checkmark$$

$$x^2 - 2x - 11 = 0 \quad \checkmark$$

$$x = \frac{2 \pm \sqrt{(-2)^2 - 4(1)(-11)}}{2(1)} \quad \checkmark$$

$$x = \frac{2 \pm \sqrt{4+44}}{2} \quad \checkmark$$

$$x = \frac{2 \pm \sqrt{48}}{2} \quad \checkmark$$

$$x = 1 + 2\sqrt{3} \text{ or } x = 1 - 2\sqrt{3} \quad \checkmark$$

iii) $\frac{1}{x^2-2x} - \frac{8}{3x^2-5x-2} = \frac{4}{3x^2+x}$ (8)

$$\frac{1}{x(x-2)} - \frac{8}{(3x+1)(x-2)} = \frac{4}{x(3x+1)} \quad \checkmark \checkmark \checkmark$$

$$3x + 1 - 8x = 4x - 8 \quad \checkmark \checkmark \checkmark$$

$$-9x = -9$$

$$x = 1 \quad \checkmark$$

Restrictions:

$$x \neq 0, x \neq -\frac{1}{3}, \text{ and } x \neq 2 \quad \checkmark$$

$$\text{iv) } x = \sqrt{x+3} - 1 \quad (6)$$

$$x + 1 = \sqrt{x+3} \quad \checkmark$$

$$(x+1)^2 = x+3 \quad \checkmark$$

$$x^2 + 2x + 1 = x + 3 \quad \checkmark$$

$$x^2 + x - 2 = 0$$

$$(x+2)(x-1) = 0 \quad \checkmark$$

$$x = -2 \text{ or } x = 1 \quad \checkmark$$

$$n/a \quad \checkmark$$

$$\text{v) } x^2 + 4x + \frac{12}{x^2+4x} = -7 \quad (8)$$

$$x^2 + 4x + \frac{12}{x^2+4x} = -7$$

$$\text{Let } k = x^2 + 4x \quad \checkmark$$

$$k + \frac{12}{k} = -7 \quad \checkmark$$

$$k^2 + 7k + 12 = 0 \quad \checkmark$$

$$(k+4)(k+3) = 0 \quad \checkmark$$

$$x^2 + 4x + 4 = 0 \text{ or } x^2 + 4x + 3 = 0 \quad \checkmark$$

$$(x+2)(x+2) = 0 \text{ or } (x+3)(x+1) = 0 \quad \checkmark$$

$$x = -2 \text{ or } x = -3 \text{ or } x = -1 \quad \checkmark$$

Restrictions:

$$x \neq 0 \text{ and } x \neq -4 \quad \checkmark$$

$$\text{vi) } x^2 + 5x - 6 < 0 \quad (4)$$

$$(x+6)(x-1) < 0 \quad \checkmark$$

$$\text{CV's: } -6 \text{ and } 1 \quad \checkmark$$



$$-6 < x < 1 \quad \checkmark$$

QUESTION 2**9 MARKS**Solve for x and y in the following equations:

$$2y - x = 1 \text{ and } y - 2x^2 + 3x + 5 = 0 \quad (9)$$

$$2y - x = 1$$

$$x = 2y - 1 \checkmark$$

$$y - 2(2y - 1)^2 + 3(2y - 1) + 5 = 0 \checkmark$$

$$y - 2(4y^2 - 4y + 1) + 6y - 3 + 5 = 0$$

$$y - 8y^2 + 8y - 2 + 6y - 3 + 5 = 0 \checkmark$$

$$-8y^2 + 15y = 0 \checkmark$$

$$-y(8y - 15) = 0 \checkmark$$

$$y = 0 \checkmark \text{ or } y = \frac{15}{8} \checkmark$$

$$\text{if } y = 0$$

$$x = 2(0) - 1$$

$$x = -1 \checkmark$$

$$\text{if } y = \frac{15}{8}$$

$$x = 2\left(\frac{15}{8}\right) - 1$$

$$x = \frac{11}{4} \checkmark$$

QUESTION 3**11 MARKS**

a) Given: $x^2 + 6x + 8 = 0$

Solve for x by completing the square

(4)

$$x^2 + 6x + 9 - 9 + 8 = 0 \quad \checkmark$$

$$(x + 3)^2 - 1 = 0 \quad \checkmark$$

$$(x + 3)^2 = 1$$

$$x + 3 = \pm 1 \quad \checkmark$$

$$x = -2 \text{ or } x = -4 \quad \checkmark$$

b) Given: $f(x) = 2x^2 - 5x - 12$

(7)

Determine the minimum value of $f(x)$ and the corresponding x value that generates this minimum value.

$$f(x) = 2\left(x^2 - \frac{5}{2}x - 6\right) \quad \checkmark$$

$$f(x) = 2\left(x^2 - \frac{5}{2}x + \frac{25}{16} - \frac{25}{16} - 6\right) \quad \checkmark$$

$$f(x) = 2\left[\left(x - \frac{5}{4}\right)^2 - 7\frac{9}{16}\right] \quad \checkmark \checkmark$$

$$f(x) = 2\left(x - \frac{5}{4}\right)^2 - 15\frac{1}{8} \quad \checkmark$$

$$\text{Minimum value of } -15\frac{1}{8} \quad \checkmark \text{ at } x = \frac{5}{4} \quad \checkmark$$

QUESTION 4**6 MARKS**

a) **WITHOUT** solving the equation, determine the nature of roots of $10x^2 + x - 1 = 0$

(4)

$$\Delta = b^2 - 4ac$$

$$\Delta = 1^2 - 4(10)(-1) \quad \checkmark$$

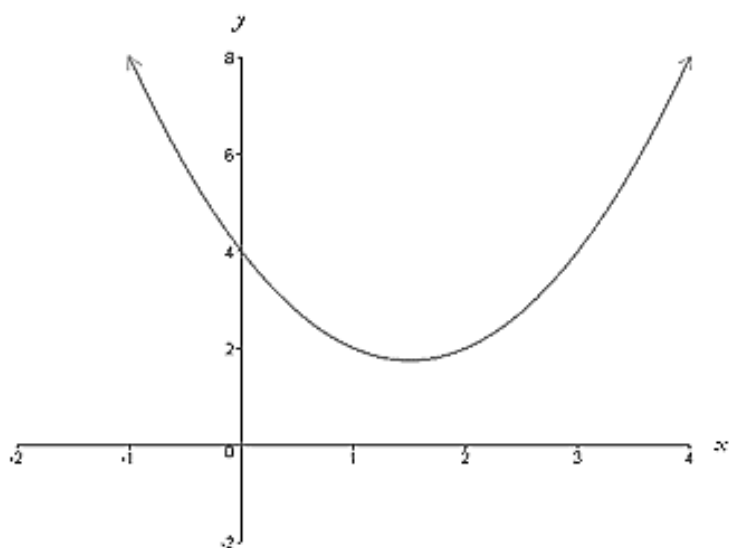
$$\Delta = 1 + 40$$

$$\Delta = 41 \quad \checkmark$$

Roots are real $\checkmark$ and irrational $\checkmark$

b) Determine the nature of roots of the function graphed below.

(2)



Roots are non-real ✓✓

QUESTION 5

8 MARKS

Steve and Paul want to ride their bicycles from school to their rugby fixture. Steve leaves school 20 min after Paul. Paul is traveling 5 km/h slower than Steve. 1 hour after Steve leaves school he catches up to Paul. How fast was Steve travelling? Complete the table below to help you. (8)

	Distance	Speed	Time
Paul	$(x - 5)(\frac{4}{3})$ ✓	$x - 5$ ✓	$\frac{4}{3}$ hours ✓
Steve	$1x$ ✓	x	1 hour

Since they rode the same distance.

$$1x = (x - 5)(\frac{4}{3}) \quad \checkmark$$

$$x = \frac{4}{3}x - \frac{20}{3}$$

$$3x = 4x - 20 \quad \checkmark$$

$$-x = -20$$

$$x = 20 \quad \checkmark$$

Steve was travelling at 20 km/h ✓